

Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Course: B.Tech

Code: 105303

Subject: Object Oriented Programming Using Java

Time: 03 Hours

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) Which of the following is not a core feature of Object-Oriented Programming?
 - (i) Inheritance
 - (ii) Encapsulation
 - (iii) Polymorphism
 - (iv) ~~Compilation~~
- (b) What is the main purpose of a class in Java?
 - (i) To create objects
 - (ii) To store data only
 - (iii) To execute the program
 - (iv) To handle exceptions
- (c) Which access modifier provides the highest visibility in Java?
 - (i) Private
 - (ii) Protected
 - (iii) Public
 - (iv) Default
- (d) Method overloading is an example of
 - (i) Compile-time polymorphism
 - (ii) Runtime polymorphism
 - (iii) Inheritance
 - (iv) Abstraction
- (e) Method overriding demonstrates
 - (i) Compile-time polymorphism
 - (ii) Runtime polymorphism
 - (iii) Encapsulation
 - (iv) Abstraction
- (f) Which of the following is used to achieve 100% abstraction in Java?
 - (i) Abstract class
 - (ii) Interface
 - (iii) Final class
 - (iv) Static class
- (g) A class can implement
 - (i) Only one interface
 - (ii) Multiple interfaces
 - (iii) Only one abstract class
 - (iv) Multiple abstract classes
- (h) The Object class is
 - (i) The root of all classes in Java
 - (ii) A user-defined class
 - (iii) An interface
 - (iv) A final class only
- (i) Which keyword is used to import a package?
 - (i) package
 - (ii) import
 - (iii) include
 - (iv) using
- (j) String class in Java is
 - (i) Mutable
 - (ii) Immutable
 - (iii) Abstract
 - (iv) Final but mutable

Q.2 (a) Explain the different types of access modifiers available in Java and describe their visibility scopes with suitable examples. [7]

(b) Explain the difference between procedural programming and object-oriented programming paradigms. Why is Java considered an object-oriented language? [7]

- Q.3** (a) Describe the concept of method overloading and method overriding with the help of suitable examples from Java. [7]
(b) Explain the difference between an interface and an abstract class. How does Java support multiple inheritance using interfaces? [7]
- Q.4** (a) Describe the exception hierarchy in Java. Differentiate between checked and unchecked exceptions with examples. [7]
(b) Write a complete Java program to demonstrate the use of constructors (default, parameterized, copy), method overloading, and the 'this' keyword. [7]
- Q.5** (a) Write a Java program to handle multiple exceptions using nested try-catch blocks and the finally clause. Include re-throwing of exceptions. [7]
(b) Write a JDBC program to connect to a database, insert a record into a "Student" table, and retrieve all records using the Collections Framework (ArrayList) to store the results. [7]
- Q.6** (a) Compare and contrast inheritance hierarchies with interface implementation. Analyze when you would prefer interfaces over abstract classes in a Java project. [7]
(b) Evaluate the impact of garbage collection and the finalize() method on Java program performance. Is it advisable to rely on finalize() for resource cleanup? Justify your answer. [7]
- Q.7** (a) "Multithreading improves performance but introduces complexity." Critically evaluate this statement with reference to thread synchronization, priorities, and inter-thread communication. [7]
(b) Explain the key features of Object-Oriented Programming and describe how each feature provides advantages over procedural programming. [7]
- Q.8** (a) What is the purpose of break and continue statements in loops? Explain with examples. [7]
(b) Write a Java program to print all odd numbers between 1 to 50 using a while loop. [7]
- Q.9** Write short note on *any two* of the following: [7 x 2 = 14]
(a) History of java
(b) Concept of Packages in Java
(c) Polymorphism
(d) Multiple inheritance
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